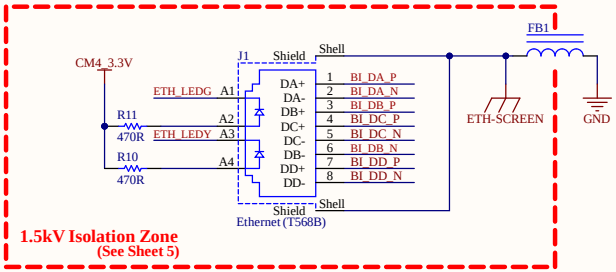
DRAWING NUMBER	8610-0533
USED ON	8660-0533
Board No.	8600-0533
	PCB

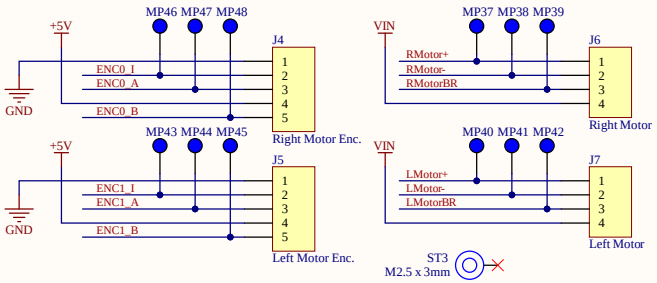
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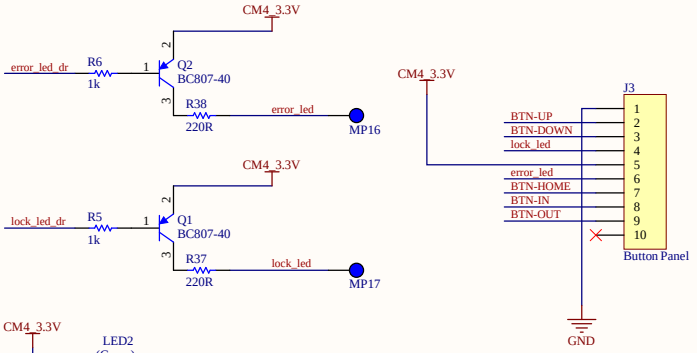
1) '*' = part not fitted

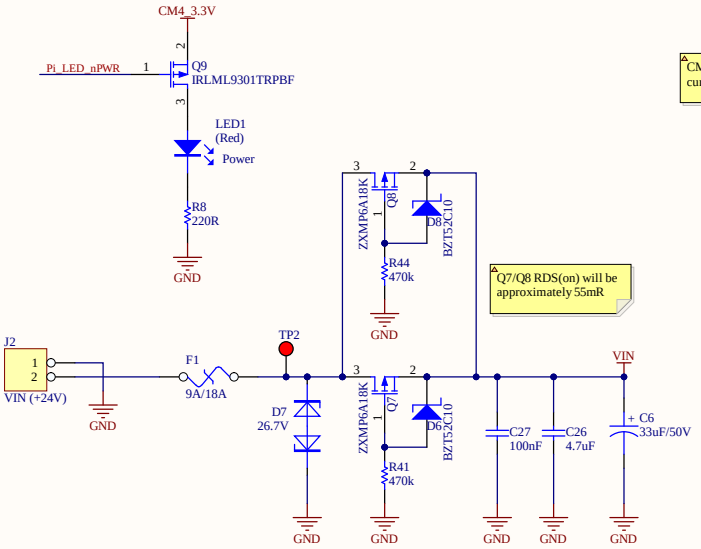
REV	DATE	DESCRIPTION	DRN	CHK	APP
1.0.0	02/02/23	Initial Issue	JC	NO	AN

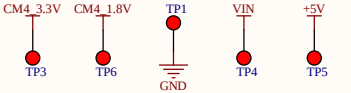


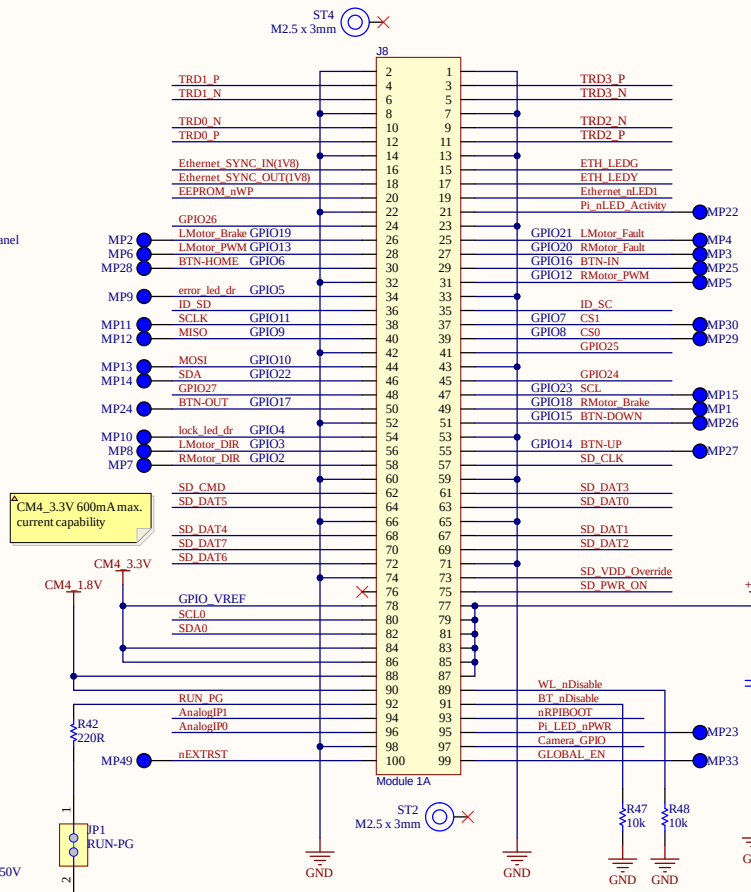
RJ45 Pin #	Wire Color (T568B)	Wire Diagram (T568B)	10Base-T Signal 100Base-TX Signal	1000Base-T Signal
1	White/Orange		Transmit+	BI_DA+
2	Orange		Transmit-	BI_DA-
3	White/Green		Receive+	BI_DB+
4	Blue		Unused	BI_DC+
5	White/Blue		Unused	BI_DC-
6	Green		Receive-	BI_DB-
7	White/Brown		Unused	BI_DD+
8	Brown		Unused	BI_DD-

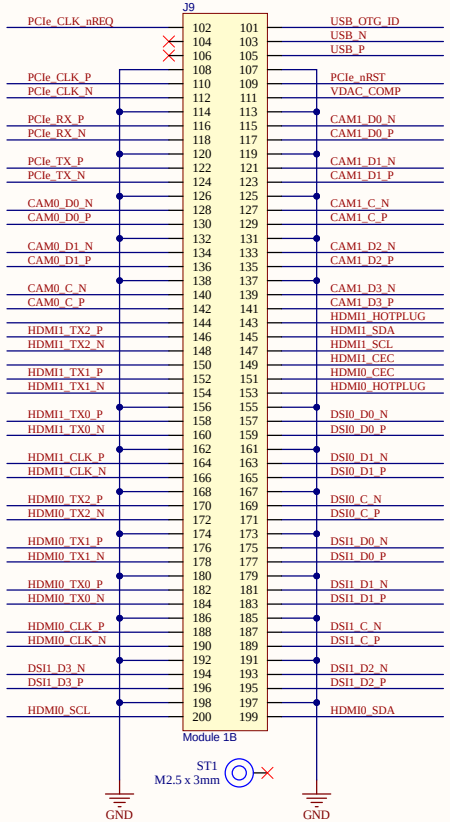












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DATE	02/02/23
CHECKED	NO
DATE	02/02/23
APPROVED	AN
DATE	02/02/23
SCALE	NOT TO SCALE
DWG STD	BS 8888

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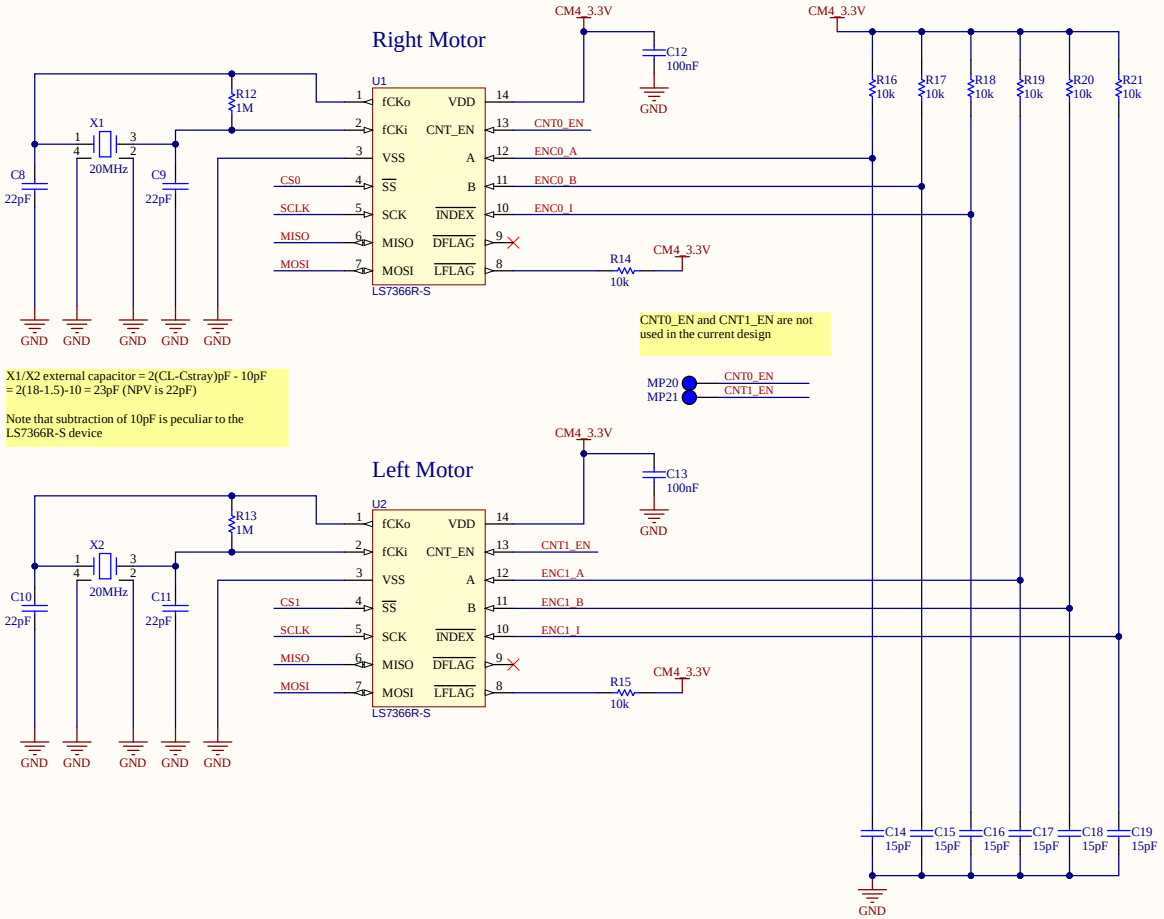
TITLE	DaWinchi Main Board Top Level
DRAWING NUMBER	8610-0533
REV	1.0.0
SHEET SIZE	A3
FILE NAME	8610-0533 Rev 1.0.0 Sht 1 - Top Level.SchDoc
SHEET	1 of 5

DRAWING NUMBER
8610-0533
USED ON
8660-0533

NOTES

1) '*' = part not fitted

REV	DATE	DESCRIPTION	DRN	CHK	APP
-	-	See Sheet 1 For Information	-	-	-



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TITLE: DaWinchi Main Board
Encoders

DRAWING NUMBER	8610-0533	REV 1.0.0	SHEET SIZE A3
FILE NAME	8610-0533 Rev 1.0.0 Sht 2 - Encoders.SchDoc	SHEET 2 of 5	

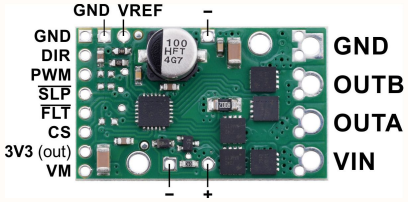
DRAWING NUMBER	
8610-0533	
USED ON	
8660-0533	

NOTES

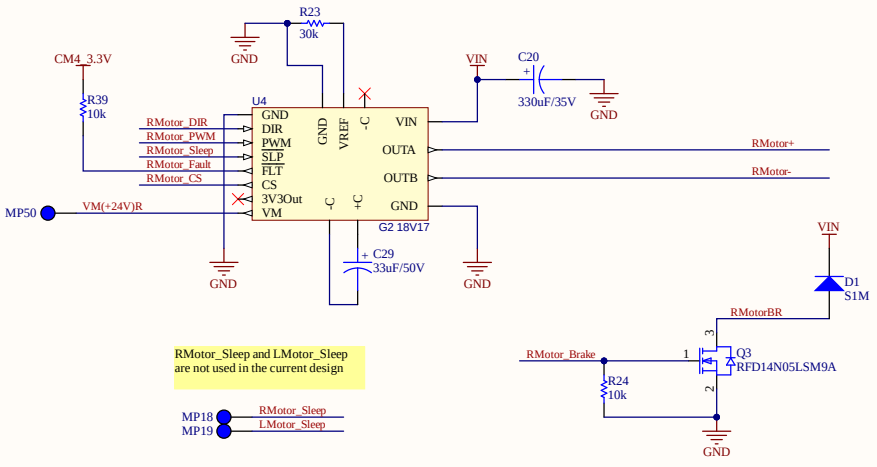
1) '*' = part not fitted

REV	DATE	DESCRIPTION	DRN	CHK	APP
-	-	See Sheet 1 For Information	-	-	-

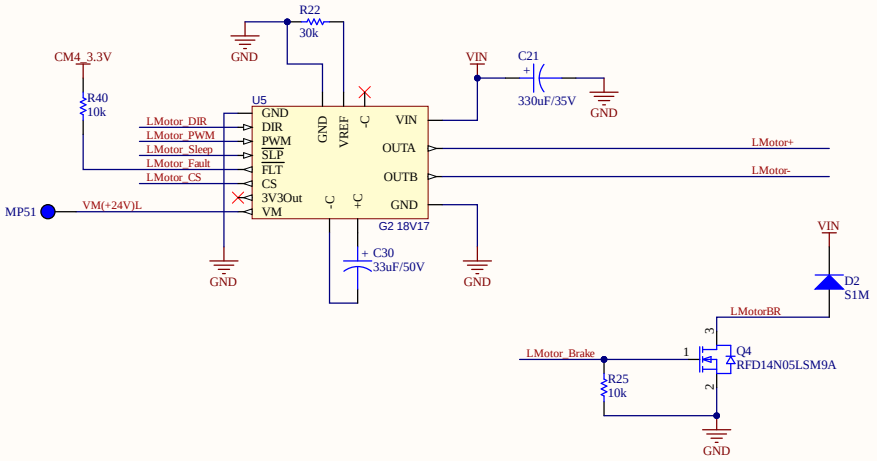
G2 High-Power Motor Driver 18V17

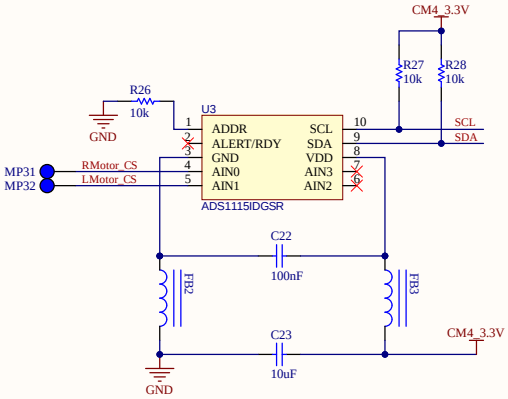


Right Motor



Left Motor





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DATE	02/02/23
SCALE	NOT TO SCALE
DWG STD	BS 8888

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TITLE: DaWinchi Main Board
Motor Driver

DRAWING NUMBER	8610-0533	REV	1.0.0
FILE NAME	8610-0533 Rev 1.0.0 Sht 3 - Motor Driver.SchDoc	SHEET	3 of 5
		SHEET SIZE	A3

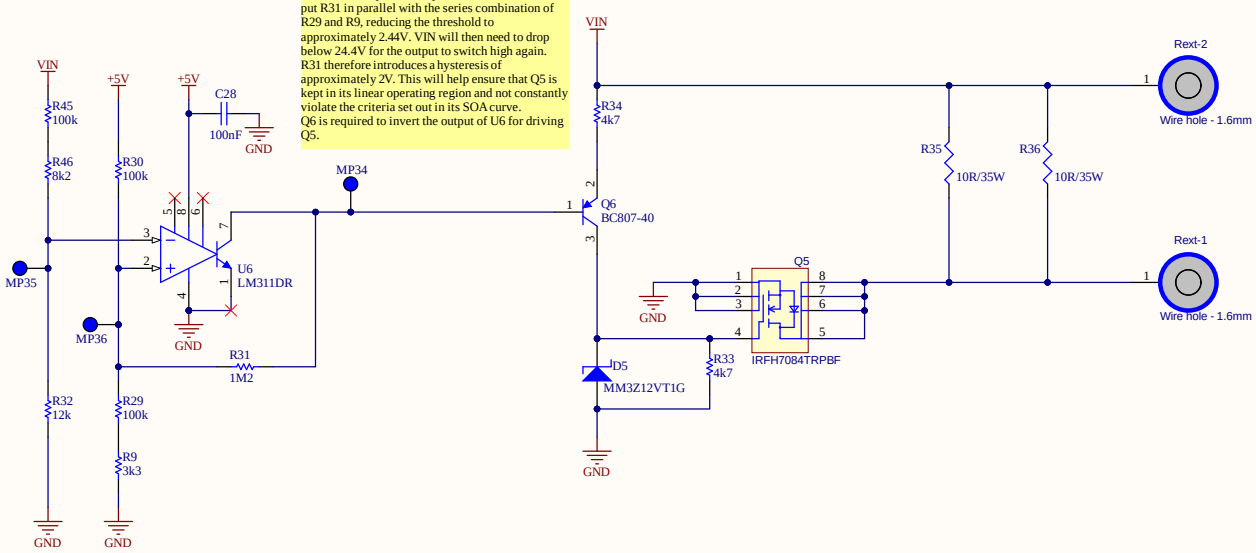
DRAWING NUMBER
8610-0530
USED ON
8660-0533

NOTES

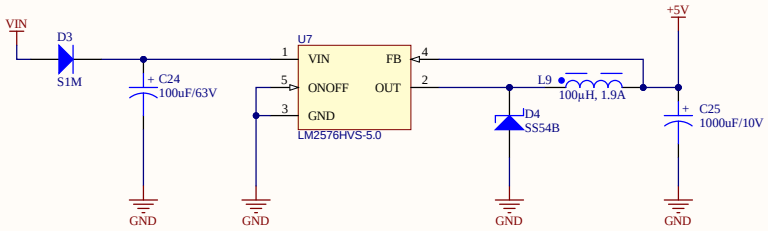
1) '*' = part not fitted

The reference voltage on the V+ input of comparator, U6, is 2.54V.
When VIN is the expected value of 24V the V- input of U6 will be 2.4V and its output will be high. This puts R31 in parallel with R30, which effectively increases the threshold to approximately 2.64V. This means that the voltage on the V- input of U6 will need to rise to approximately 26.4V for its output to switch low. If VIN increases above 26.4V the comparator output will switch low and put R31 in parallel with the series combination of R29 and R9, reducing the threshold to approximately 2.44V. VIN will then need to drop below 24.4V for the output to switch high again. R31 therefore introduces a hysteresis of approximately 2V. This will help ensure that Q5 is kept in its linear operating region and not constantly violate the criteria set out in its SOA curve.
Q6 is required to invert the output of U6 for driving Q5.

REV	DATE	DESCRIPTION	DRN	CHK	APP
-	-	See Sheet 1 For Information	-	-	-



Rext-1 and Rext-2 are for fitting a shunt resistor (additional to R35 and R36) to absorb excess motor power in the event that one or both of the motors are attempting to feed power back to the supply (a typical scenario when motors are slowing down).



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DWG STD	BS 8888

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TITLE: DaWinchi Main Board
Power

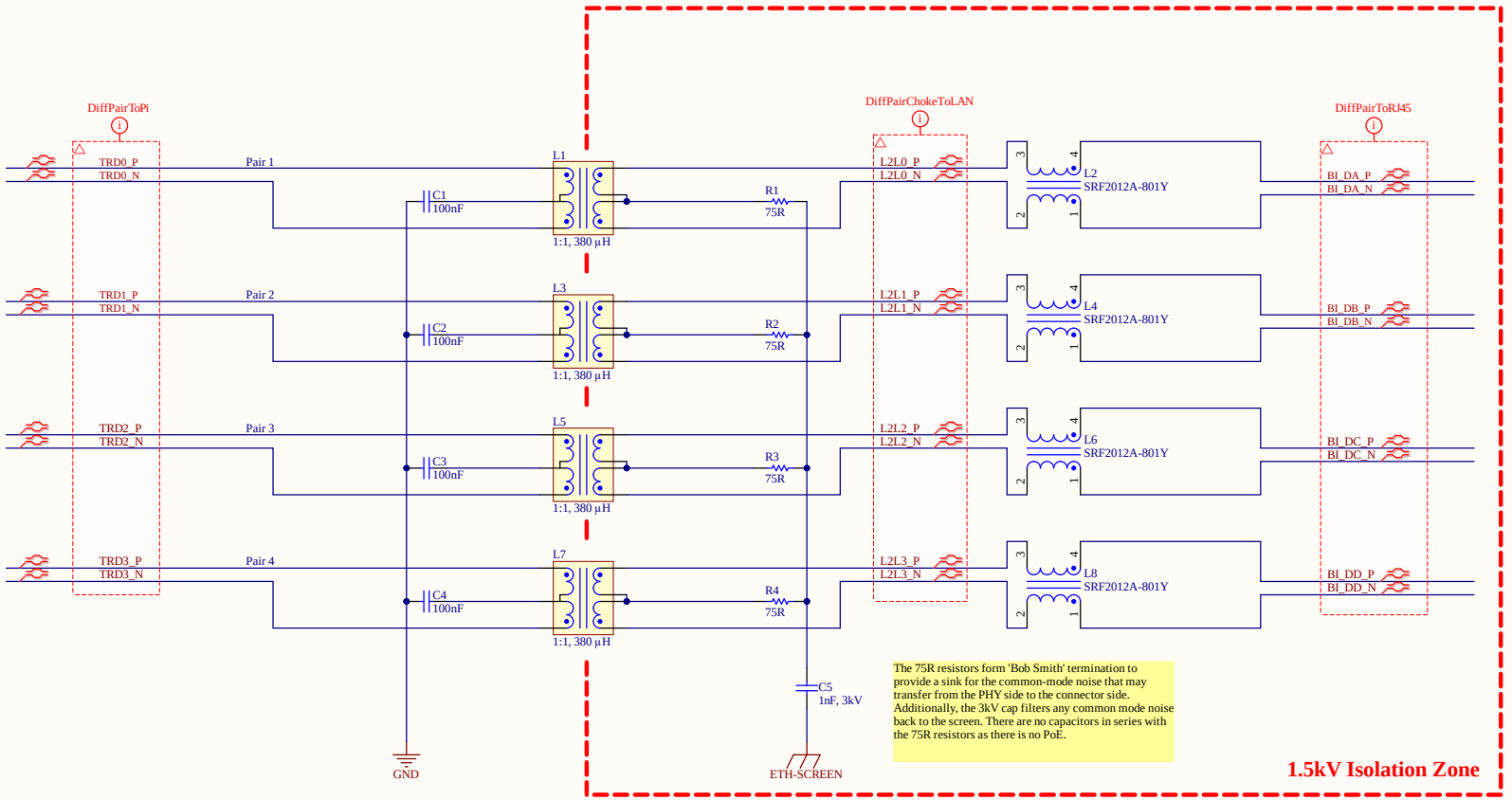
DRAWING NUMBER	8610-0530	REV	1.0.0
FILE NAME	8610-0533 Rev 1.0.0 Sht 4 -Power.SchDoc	SHEET	4 of 5
		SHEET SIZE	A3

DRAWING NUMBER
8610-0533
USED ON
8660-0533

NOTES

1) '*' = part not fitted
2) Match pair length < 0.1mm across the PCB
3) Match Group length < 0.5mm across the PCB
4) Differential pair impedance = 100R
5) Avoid stubs
6) 1.5kV isolation zone: maintain 0.5mm clearance and 5.6mm creepage distance as EN62368-1 (pollution class 1) where practical (not under transformers due to pin pitch)

REV	DATE	DESCRIPTION	DRN	CHK	APP
-	-	See Sheet 1 For Information	-	-	-



The 75R resistors form 'Bob Smith' termination to provide a sink for the common-mode noise that may transfer from the PHY side to the connector side. Additionally, the 3kV cap filters any common mode noise back to the screen. There are no capacitors in series with the 75R resistors as there is no PoE.

DIFF PAIR LENGTHS TO RASPBERRY PI

PAIR 1
1621.398mils
1621.397mils

PAIR 2
1621.395mils
1621.396mils

PAIR 3
1621.396mils
1621.396mils

PAIR 4
1621.396mils
1621.396mils

Maximum diff pair length difference is with pair 1 (or 2) = 1621.398-1621-397 (or 1621.396-1621-395) = 0.001mils = 0.001*25.4/1000 = 0.000025mm approx.
Maximum group length difference = 1621.398-1621.385 = 0.003mils = 0.003*25.4/1000 = 0.000076mm approx.

DIFF PAIR LENGTHS BETWEEN XFRMRS AND FILTERS

PAIRS 1 - 4
120.657mils
120.656mils

Maximum diff pair length difference = 120.657-120.656 = 0.001mils = 0.001*25.4/1000 = 0.000025mm approx.
Maximum group length difference = 0mils = 0mm.

DIFF PAIR LENGTHS TO RJ45

PAIR 1
615.275mils
615.275mils

PAIR 2
615.275mils
615.275mils

PAIR 3
615.275mils
615.275mils

PAIR 4
615.275mils
615.275mils

Maximum diff pair length difference #NAME? = 0mm.
Maximum group length difference = 0mils = 0mm.

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DWG STD	BS 8888

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TITLE: DaWinchi Main Board Ethernet

DRAWING NUMBER	8610-0533	REV 1.0.0	SHEET SIZE A3
FILE NAME	8610-0533 Rev 1.0.0 Sht 5 - Ethernet.SchDoc	SHEET 5 of 5	